THE IMPACT OF NON-STANDARD BARLEY QUALITY ON THE MALTING PROCESS AND ITS UTILIZATION POSSIBILITIES

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INTRODUCTION

Barley is one of the most important agricultural crops worldwide with a wide range of uses and is particularly important in the feed and malting industries. The quality of malting barley is influenced by many factors during cultivation, such as variety, climatic conditions, soil and fertilisation, but also by post-harvest factors such as storage. However, the final product - malt - is also influenced by the technology of its production. By modifying some technological steps (cleaning), it is possible to increase the quality of the barley itself, to promote its germination (addition of enzymes), or to a large extent to modify the malting technology so that the malt obtained is of higher quality despite the lower quality of malting barley (modification of steeping, air rest, modification of kilning, etc.). Barley of non-standard quality may be defined as grain with unsatisfactory moisture content, insufficient or high nitrogen content, biological or mechanical deterioration, mycotoxin or heavy metal content and poor/irregular germination. In view of the hygienic quality of the malt produced, barley with exceeded values of the parameters monitored in relation to possible negative effects on the health of the consumer cannot be used and the parameters in question must therefore be strictly always observed. Barley of non-standard quality is rarely processed into malt and results in inferior quality malt that can adversely affect the sensory, chemical and physical properties of beer. Batches of poor quality barley are most commonly processed as: livestock feed, raw material for the production of a products other than malt, e.g. barley flour, barley extrudate, raw material for biomass or bioethanol production, conversion into fish feed.

The aim of this work was to analyse non-standard malting barley technological quality and to determine the malting potential of produced malt samples and its utilization possibilities..

MATERIAL AND METHODS

Micro-malting was conducted in a laboratory micro-malting plant (Ravoz, Olomouc, Czech Republic) provided by Research Institute AgroBioTech at Slovak Agricultural University in Nitra. All determinations were carried out according to European Brewery Convention recommended methods (EBC, 2010) and the Middle European Brewing Analysis Commission methods (MEBAK, 2011).

Micro-malting

Barley and malt analyses

Table 2 Evaluation of malt samples technological parameters

Malting parameters	Casanova A	Casanova B	Casanova C	Casanova D	Casanova E	Overture A	Overture B	Overture C
Crude protein (%)	13.68±0.06f	7.74±0.05a	8.39±0.02b	12.52±0.40e	9.34±0.28d	12.69±0.22e	8.49±0.09b	8.92±0.12c
Extract of malt (%)	77.69±0.37a	83.81±0.25ef	82.88±0.26cd	80.28±0.55b	83.07±0.41de	82.26±0.51c	84.12±0.63f	82.86±0.48cd
Kolbach index (%)	25.55±0.54a	40.87±0.28f	41.55±0.62f	29.63±0.33b	36.58±0.41d	32.83±1.08c	38.63±0.63e	30.71±0.97b
Friability (%)	55.56±0.52a	85.96±0.92e	79.68±1.05d	61.28±0.32b	85.00±0.19e	78.04±0.36c	97.54±0.46f	80.10±0.49d
Relative extract 45 °C (%)	29.29±0.72b	27.59±0.43a	27.92±0.78a	32.79±0.43e	31.71±0.53c	37.30±0.60f	30.28±0.45b	31.80±0.63cd
Wort viscosity (mPa.s^{-1})	1.56±0.00e	1.51±0.00c	1.53±0.01d	1.67±0.01f	1.47±0.01b	1.75±0.00g	1.44±0.01a	1.55±0.01e
Wort haze 90° (j. EBC)	4.15±0.04b	8.48±0.05g	7.34±0.05d	7.48±0.02e	7.03±0.10c	3.48±0.03a	3.58±0.08a	8.11±0.06f
Wort colour (j. EBC)	5.03±0.05a	8.03±0.06h	7.14±0.04d	7.61±0.04e	7.85±0.04g	5.75±0.08b	5.93±0.04c	7.72±0.05f
Apparent final attenuation (%)	81.70±0.12b	81.91±0.49bc	81.90±0.32bc	80.62±0.36a	80.83±0.21a	82.83±0.11d	82.62±0.58cd	80.31±0.73a
Filtration time (min.)	18.33±1.53bc	17.67±0.58b	18.33±1.53bc	14.67±0.58a	20.33±1.53c	19.33±1.53bc	39.33±1.53e	31.33±1.53d

Table 1 Evaluation of barley samples technological parameters

Barley parameters	Casanova A	Casanova B	Casanova C	Casanova D	Casanova E	Overture A	Overture B	Overture C
Crude protein (%)	13.93 ±0.08g	8.54 ±0.09a	9.02 ±0.04b	12.74±0.06e	9.94±0.08 d	13.66±0.17f	9.33±0.09c	9.38±0.11c
Starch (%)	53.19±0.88.a	58.48±0.75e	57.97±0.7de	54.71±0.16b	57.11±0.13c	53.05±0.38a	57.61±0.32cd	57.63±0.45cd
First class proportion (%)	65.53±0.55a	95.10±0.53e	94.00±1.06cd	98.20±0.60f	97.97±0.35f	93.57±1.29c	98.37±0.95f	89.77±0.95b
Hectoliter weight (g.dm $^{-3}$)	677.23±4.47bc	679.93±3.00bc	670.07±3.05ab	733.23±10.55d	684.33±5.85c	681.67±3.98c	660.40±9.75a	687.20±5.37c
Germination (%)	98.67±0.58a	99.00±0.00ab	99.67±0.58bc	98.67±0.58a	100.00±00c	99.67±0.58bc	98.50±0.50a	98.50±0.50a
Steeping degree - end of steeping (%)	39.67±0.58c	36.33±1.15a	39.67±0.58c	36.33±0.58a	40.33±1.15c	39.00±1.00bc	38.00±1.00b	36.33±0.58b
Steeping degree - first day of germ. (%)	42.33±1.15c	39.33±0.58ab	40.33±1.53abc	39.00±1.00a	41.33±1.53bc	40.67±0.58abc	41.33±1.53bc	38.67±1.53a
Steeping degree - end of germination (%)	38.33±1.53a	38.33±0.58a	38.67±1.53a	37.33±0.58a	37.33±1.15a	37.33±1.53a	38.33±0.58a	39.33±1.53a

CONCLUSION

According to the obtained results, it can be concluded, that malting of barley with non-standard quality in terms of nitrogenous substances may affect the water uptake by the grain, but also the drying of the malt and therefore the resulting moisture content of the malt. The high nitrogen content of barley negatively affected grain modification, malt extract and Kolbach index. For this reason, it is necessary to use mainly barley of optimal technological quality for malting purposes. In case of the use of barley of non-standard quality, an individual approach to the malting process is necessary by ensuring the optimum degree of steeping of the barley, as well as the correct adjustment of the malting technological process of germination and kilning. Blending can significantly improve the quality of malt and may involve blending malts of different qualities, for example, blending malts with different friability values, nitrogen-values, apparent final attenuation in order to achieve the specification of the final malt according to the customer's requirements.

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